

航线管理

📅 2025-11-12 ★ ★ ★ ★ ★ 4.6 分 3 用户已评分

[Github Edit](#)

- 返航信息
- 任务就绪通知
- 上报航线任务进度
- 设备返航退出状态通知
- 空中下发航线状态上报
- 执行任务
- 指定home点的返航
- 任务终止
- 取消返航
- 一键返航

Event

返航信息

Topic: thing/product/{gateway_sn}/events

Direction: up

Method: return_home_info

Data:

Column	Name	Type	constraint
planned_path_points	规划的轨迹点列表	array	{"size": -, "item_type": stru
»latitude	轨迹点纬度(角度值)	double	{"max":90,"min":-90}

»longitude	轨迹点经度(角度值)	double	{"max":180,"min":-180}
»height	轨迹点高度	float	{"max":"","min":"","step":0}
last_point_type	返航路径最后一个点的类型	enum_int	{"0":"轨迹最后一个点在返后一个点不在返航点上空"
flight_id	任务ID	text	
home_dock_sn	home点机场sn	text	
multi_dock_home_info	蛙跳任务机场返航信息	array	{"size": 2, "item_type": stru
»sn	机场SN	text	
»plan_status	路径规划状态	enum_int	{"0":"规划失败或正在规划达","2":"规划路径因电量不

»estimated_battery_consumption	预估 电量 消耗	int	{"max":"100","min":"0","sta
»home_distance	home 点距 离	float	{"unit_name":"米 / m"}

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "flight_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "home_dock_sn": "6QCDL870020164",
    "last_point_type": 0,
    "multi_dock_home_info": [
      {
        "estimated_battery_consumption": 30,
        "home_distance": 298.3277282714844,
        "plan_status": 3,
        "sn": "6QCDL870020164"
      },
      {
        "estimated_battery_consumption": 30,
        "home_distance": 298.289794921875,
        "plan_status": 3,
        "sn": "6QCDL820020041"
      }
    ],
    "planned_path_points": [
      {
        "height": 60.285194396972656,
        "latitude": 22.755022128112614,
        "longitude": 114.89828051067889
      },
      {
        "height": 88.22719192504883,
        "latitude": 22.755022128112614,
        "longitude": 114.89828051067889
      },
      {
        "height": 88.22719192504883,
        "latitude": 22.75721542071551,
        "longitude": 114.89853624254465
      },
      {
        "height": 88.22719192504883,
        "latitude": 22.757445462048054,
```

```
        "height": 78.30619430541992,  
        "latitude": 22.757531544193625,  
        "longitude": 114.89862810820341  
      }  
    ]  
  },  
  "gateway": "6QCDL820020041",  
  "method": "return_home_info",  
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",  
  "timestamp": 1720000266772  
}
```

任务就绪通知

Topic: thing/product/{gateway_sn}/events

Direction: up

Method: flighttask_ready

Data:

Column	Name	Type	constraint	Description
flight_ids	计划 ID	array	{"size": -, "item_type": text}	当前满足任务就绪条件的 任务 ID 集合

Example:

```
{  
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",  
  "data": {  
    "flight_ids": [  
      "aaaaaaa",  
      "bbbbbbb"  
    ]  
  },  
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",  
  "timestamp": 1654070968655,  
  "method": "flighttask_ready"  
}
```

上报航线任务进度

Direction: up

Method: flighttask_progress

Data:

Column	Name	Type	constraint
ext	扩展内容	struct	
»current_waypoint_index	当前执行到的航点数	int	
»wayline_mission_state	航线任务状态	enum_int	{ "0": "断连", "1": "不支持该航点", "2": "可上传文件,可执行已有文件", "3": "任务中", "4": "触发开始命令, 飞行器编辑, 还未开始任务, 处于准备中", "5": "到达第一个航点", "6": "航线执行", "7": "任务完成", "8": "任务失败", "9": "任务取消", "10": "任务暂停", "11": "任务恢复", "12": "任务结束", "13": "任务异常", "14": "任务异常", "15": "任务异常", "16": "任务异常", "17": "任务异常", "18": "任务异常", "19": "任务异常", "20": "任务异常", "21": "任务异常", "22": "任务异常", "23": "任务异常", "24": "任务异常", "25": "任务异常", "26": "任务异常", "27": "任务异常", "28": "任务异常", "29": "任务异常", "30": "任务异常", "31": "任务异常", "32": "任务异常", "33": "任务异常", "34": "任务异常", "35": "任务异常", "36": "任务异常", "37": "任务异常", "38": "任务异常", "39": "任务异常", "40": "任务异常", "41": "任务异常", "42": "任务异常", "43": "任务异常", "44": "任务异常", "45": "任务异常", "46": "任务异常", "47": "任务异常", "48": "任务异常", "49": "任务异常", "50": "任务异常", "51": "任务异常", "52": "任务异常", "53": "任务异常", "54": "任务异常", "55": "任务异常", "56": "任务异常", "57": "任务异常", "58": "任务异常", "59": "任务异常", "60": "任务异常", "61": "任务异常", "62": "任务异常", "63": "任务异常", "64": "任务异常", "65": "任务异常", "66": "任务异常", "67": "任务异常", "68": "任务异常", "69": "任务异常", "70": "任务异常", "71": "任务异常", "72": "任务异常", "73": "任务异常", "74": "任务异常", "75": "任务异常", "76": "任务异常", "77": "任务异常", "78": "任务异常", "79": "任务异常", "80": "任务异常", "81": "任务异常", "82": "任务异常", "83": "任务异常", "84": "任务异常", "85": "任务异常", "86": "任务异常", "87": "任务异常", "88": "任务异常", "89": "任务异常", "90": "任务异常", "91": "任务异常", "92": "任务异常", "93": "任务异常", "94": "任务异常", "95": "任务异常", "96": "任务异常", "97": "任务异常", "98": "任务异常", "99": "任务异常" }
»media_count	本次航线任务执行产生的媒体文件数量	int	
»track_id	航迹ID	text	
»flight_id	任务ID	text	
»break_point	航线断点信息	struct	

»»index	断点序号	int	
»»state	断点状态	enum_int	{"0":"在航段上","1":"在航点上"}
»»progress	当前航段进度	float	{"max":"1.0","min":"0","step":"","
»»wayline_id	航线ID	int	
»»break_reason	中断原因	enum_int	{"0":"无异常","1":"Mission ID 未执行","2":"不常见错误，建议持","4":"请求开始/恢复航线任务出错，请重新尝试上传文件开始持","5":"请求查询断点信息时，失败。请求恢复航线任务时，解析失败","6":"请求开始/结束航线任务误，协议请求的指令有误。请习解析断点类型失败","7":"请求开时，解析 wpmz 文件超时，请重经开始，不能再次开始","258":"航线，只允许在航线执行状态时线","259":"航线未开始，不能结任务冲突，无法获取飞行器控制和返航中开始航线","262":"该状线，只允许在航线暂停状态时恢飞行器超过限高高度","514":"飞行离","515":"航线穿过限飞区，机务","516":"飞行器触发限低","51障","518":"RTK 信号差","519":"挂界","521":"超过机场限飞区限高飞失败","523":"起飞任务执行失任务失败","526":"请求航线 RTK 败","527":"航线 RTK 收敛任务运障碍物或者禁飞区域，导致航线达","769":"GPS 信号弱","770":"i 档","771":"返航点未刷新","772"开始任务","773":"低电量返航导断","775":"遥控器与飞行器失联面起桨，不允许开始航线","779中，相机状态异常（如过亮，达致）","780":"实时仿地用户设置（大于 200 m 或者小于 30 m）过程中全局地图计算出错","784

			断","1283":"用户触发返航","153点 index 或者 progress) 错误","的坐标系","1541":"使用不支持的模式","1542":"使用不支持的过渡模式","1543":"使用不支持的 yaw 不支持的 yaw 方向调转模式","1的航点类型","1546":"首尾点不能型","1547":"航线全局速度超过范围","1548":"航点数量异常","154常","1550":"转弯截距异常","155超过合理范围","1552":"航段目标范围","1553":"航点 yaw 角度超过围","1555":"断点续飞的 mission 误","1556":"断点续飞的 progress 误","1557":"断点续飞的任务状态点续飞的航点 index 信息输入错误飞的经纬度信息输入错误","156 yaw 输入错误","1561":"断点续误","1563":"航线生成失败","156败","1565":"航线避障紧急刹停"的动作类型","1595":"动作 ID 不作 ID 值不能为 65535","1602":"理范围","1603":"动作组生效范围点续飞中动作 index 超过合理范息中触发器运行结果异常","160组 ID 信息不能重复","1610":"断置不能重复","1611":"断点续飞理范围","1612":"续飞中动作 ID 中","1613":"断点续飞中不能修断","1614":"断点信息错误导致失败","1634":"无法识别的动作类型别的触发器类型","65534":"未知误","65535":"未知内部错误")
»»latitude	断点 纬度	float	{"max":"90","min":"-90","step":"°"}
»»longitude	断点 经度	float	{"max":"180","min":"-180","step":"/°"}
»»height	断点 相对地球 椭球 面高 度	float	{"unit_name":"米 / m"}

»»attitude_head	断点 偏航 轴角 度	float	
»wayline_id	当前 在作 业的 航线 ID	int	
status	任务 状态	enum_string	{ "canceled": "取消或终止", "failed": "失败", "in_progress": "执行中", "ok": "成功", "partially_done": "部分完成", "stopped": "停止", "rejected": "拒绝", "sent": "已发送" }
progress	进度	struct	
»current_step	执行 步骤	enum_int	{ "0": "初始状态", "1": "启动前检查", "2": "启动前检查, 模式", "3": "启动前检查, 航线执行", "4": "航线执行, 返航中", "5": "航线执行进入待任务下发", "6": "机场进入工作检查准备工作和开盖准备工作", "7": "开盖", "8": "起飞", "9": "等待飞行系统准备就绪", "10": "等待 RTK 源监听有值", "11": "RTK 源是否是机场源, 如果不是置", "12": "等待飞行控制权通知", "13": "飞行控制权, 抢夺飞行器控制权", "14": "自检", "15": "离线地图一致性检查", "16": "下载 KMZ URL", "17": "下载 KMZ", "18": "返航中", "19": "染色配置", "20": "飞行器降点设置, 起飞高度设置, 染色", "21": "flyto 起飞参数设置", "22": "起飞准备状态", "23": "Home 点设置", "24": "航线", "25": "进入返航的检查准备", "26": "降落机场", "27": "降落以后的关盖", "28": "模式", "29": "机场异常恢复", "30": "日志", "31": "相机录像状态检查", "32": "数量", "33": "机场起飞开盖的异常", "34": "任务结果", "35": "日志列表拉取 - 7", "36": "日志列表拉取 - 拉取机场列表", "37": "上传日志列表结果", "38": "日志拉"


```
压缩飞行器日志","42":"日志拉取  
志","43":"日志拉取-上传飞行器  
取-上传机场日志","45":"日志拉  
果","46":"自定义飞行区文件更  
义飞行区更新中","48":"离线地图  
中","49":"离线地图更新中","655  
务回包","65534":"无具体状  
态","65535":"UNKNOWN"}

```

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "output": {
      "ext": {
        "break_point": {
          "attitude_head": 30,
          "break_reason": 1,
          "height": 100.23,
          "index": 1,
          "latitude": 23.4,
          "longitude": 113.99,
          "progress": 0.34,
          "state": 0,
          "wayline_id": 0
        },
        "current_waypoint_index": 3,
        "flight_id": "flight_id",
        "media_count": 6,
        "track_id": "track_id",
        "wayline_id": 0,
        "wayline_mission_state": 9
      },
      "progress": {
        "current_step": 19,
        "percent": 100
      },
      "status": "ok"
    },
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
}
```

设备返航退出状态通知

Topic: thing/product/{gateway_sn}/events

Direction: up

Method: device_exit_homing_notify

Data:

Column	Name	Type	constraint	Description
sn	机场 SN	text	{}	
action	退出返航通知消息类型	enum_int	{"0":"退出"返航退出状态","1":"进入"返航退出状态"}	进入“返航退出状态”，指当机场处于返航模式时，由于 <code>reason</code> 字段中展示的原因，退出了返航过程。相似的，退出“返航退出状态”，指机场停止了退出返航这一过程。
reason	退出返航原因	enum_int	{"0":"操纵杆油门添加","1":"操纵杆间距添加","2":"行为树初始化失败","3":"被障碍物包围","4":"触发限飞限制","5":"障碍物距离太近","6":"无 GPS 信号","7":"GPS 和 VIO 位置输出标志为 false","8":"GPS 和 VIO 融合位置误差太大","9":"短距离回溯","10":"近距离触发返航"}	

Example:

```
{
  "data": {
    "action": 1,
    "reason": "0",
    "sn": "机场 SN"
  },
  "need_reply": 1,
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "device_exit_homing_notify"
}
```

空中下发航线状态上报

Method: `in_flight_wayline_progress`

Topic: `thing/product/{gateway_sn}/events`

Direction: up

Data:

Column	Name	Type	Constraint	Description
<code>in_flight_wayline_id</code>	航线任务ID	string		空中航线任务的唯一标识符
<code>progress</code>	任务进度	object		任务进度信息
<code>progress.percent</code>	完成百分比	int	<code>{min: 0, max: 100}</code>	任务完成百分比
<code>status</code>	任务状态码	int		状态码说明: <code>status : 1</code> : 上传文件中 (<code>wayline_uploading</code>) <code>2</code> : 文件上传成功 (<code>wayline_uploaded</code>) <code>3</code> : 执行任务中 (<code>wayline_progress</code>) <code>4</code> : 任务暂停 (<code>wayline_paused</code>) <code>5</code> : 任务取消 (<code>wayline_cancel</code>) <code>6</code> : 任

				失败 (wayline_failed) 任务失败时的错误码 8 : 任务超时
result	错误原因码	int		
way_point_index	当前航点索引	int		当前执行的航点索引

Example:

```
{
  "in_flight_wayline_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "progress": {
    "percent": 100
  },
  "status": 0,
  "result": 0,
  "way_point_index": 0
}
```

Service

执行任务

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: flighttask_execute



Data:

Column	Name	Type	constraint
flight_id	计划ID	text	

multi_dock_task	蛙跳任务参数	struct	
»wireless_link_topo	图传连接拓扑	struct	
»»secret_code	加密编码	array	{"size": 28, "item_type": int}
»»center_node	飞行器对频信息	struct	
»»»sdr_id	扰码信息	int	
»»»sn	设备sn	text	
»»»leaf_nodes	机场或遥控器对频信息, 组装执行蛙跳任务的对应机场属性	array	{"size": -, "item_type": struct}
»»»sdr_id	扰码信息	int	
»»»sn	设备sn	text	
»»»control_source_index	控制源序号	int	{"max": "2", "min": "1", "step": "1",

»dock_infos	机场信息	array	{"size": 2, "item_type": struct}
»»dock_type	机场在蛙跳任务中的角色	enum_string	{"landing":"降落","takeoff":"起
»»latitude	纬度	double	{"max":"90","min":"-90","step":'
»»longitude	经度	double	{"max":"180","min":"-180","step
»»height	椭球高度	double	{"unit_name":"米 / m"}
»»heading	机场朝向角	double	{"max":"180","min":"-180","step / °"}
»»home_position_is_valid	机场home点有效性	enum_int	{"0":"无效","1":"有效"}
»»index	机场任务唯一	int	{"max":"31","min":"1"}

	识别		
»»sn	机场 SN	text	
»»rtcm_info	机场 RTK标 定源	struct	
»»»mount_point	网络 RTK挂 载点 信息	text	
»»»port	网络 端口 信息	text	
»»»host	网络 host 信息	text	
»»»rtcm_device_type	设备 类型	enum_int	{"1":"机场"}
»»»source_type	标定 类型	enum_int	{"0":"未标定","1":"自收敛标定", 定","3":"网络RTK标定"}
»»alternate_land_point	备降 点	struct	
»»»longitude	经度	float	{}
»»»latitude	纬度	float	{}
»»»height	椭球 高度	float	{}
»»»safe_land_height	安全 高度 (备降 转移 高)	float	{}
»»»is_configured	是否 设置	enum_int	{"0":"未设置","1":"已设置"}

点

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "flight_id": "aee739dc-f1cc-47d5-beaa-64d327f2d797",
    "multi_dock_task": {
      "dock_infos": [
        {
          "alternate_land_point": {
            "height": 0,
            "is_configured": 1,
            "latitude": 37.348792490321514,
            "longitude": 116.52867090782102,
            "safe_land_height": 30
          },
          "dock_type": "takeoff",
          "heading": -78.7066650390625,
          "height": 30.81130027770996,
          "home_position_is_valid": 1,
          "index": 1,
          "latitude": 37.3487657248638,
          "longitude": 116.52893686422743,
          "rtcm_info": {
            "host": "120.253.226.97",
            "mount_point": "RTCM33_GRCEJ",
            "port": "8002",
            "rtcm_device_type": 1,
            "source_type": 3
          },
          "sn": "7CTDM5900B3X1B"
        },
        {
          "alternate_land_point": {
            "height": 0,
            "is_configured": 1,
            "latitude": 37.336024417709915,
            "longitude": 116.554075635852,
            "safe_land_height": 30
          },
          "dock_type": "landing",
          "heading": -69.79183197021484,
          "height": 32.31420135498047,
          "home_position_is_valid": 1,
          "index": 2,
          "latitude": 37.33605462094235,
          "longitude": 116.55416413516038,

```



```

        "port": "8002",
        "rtcm_device_type": 1,
        "source_type": 3
    },
    "sn": "7CTDM5900BK07M"
}
],
"wireless_link_topo": {
    "center_node": {
        "sdr_id": 933765657,
        "sn": "1581F6Q8D245P00EKS87"
    },
    "leaf_nodes": [
        {
            "control_source_index": 1,
            "sdr_id": 920128532,
            "sn": "7CTDM5900B3X1B"
        },
        {
            "control_source_index": 2,
            "sdr_id": 911741468,
            "sn": "7CTDM5900BK07M"
        }
    ],
    "secret_code": [
        0,
        0,
        0,
        0,
        1,
        0,
        0,
        0,
        123,
        114,
        19,
        203,
        192,
        100,
        244,
        160,
        146,
        228,
        196,
        213,
        105,
        220,
        176,
        147,
        87,
        182,
        90,
    ]
}

```

```
    },  
    "method": "flighttask_execute",  
    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
    "timestamp": 1720095314016  
  }  
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: flighttask_execute

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{  
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "data": {  
    "result": 0  
  },  
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "timestamp": 1234567890123,  
  "method": "flighttask_execute"  
}
```

指定home点的返航

蛙跳任务中，需要指定一个机场作为返航目的地时使用

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: return_specific_home

Data:

home_dock_sn	返航的目的机场SN	text		
--------------	-----------	------	--	--

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "home_dock_sn": "xxxxxxxx"
  },
  "method": "return_specific_home",
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1234567890123
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: return_specific_home

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "method": "return_specific_home",
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1695634358385
}
```

任务终止

蛙跳任务中，当一个机场上报任务结束后，需要下发此指令到另一机场结束任务

Topic: thing/product/{gateway_sn}/services

Method: flighttask_stop

Data:

Column	Name	Type	constraint	Description
flight_id	任务id	text		
reason	结束原因	enum_int	{"0":"正常结束","1":"另一机场状态机异常"}	

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "flight_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "reason": 0
  },
  "method": "flighttask_stop",
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1234567890123
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: flighttask_stop

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "method": "flighttask_stop",

```

取消返航

返航后，飞行器会退出航线模式，此时取消返航，飞行器会悬停

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: return_home_cancel

Data: null

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {},
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "return_home_cancel"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: return_home_cancel

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "need_reply": 1,
}
```

```
}

```

一键返航

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: return_home

Data: null

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: return_home

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误
output	输出	struct		
»status	任务状态	enum_string	{ "canceled": "取消或终止", "failed": "失败", "in_progress": "执行中", "ok": "执行成功", "paused": "暂停", "rejected": "拒绝", "sent": "已下发", "timeout": "超时" }	

航线恢复

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: flighttask_recovery

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {},
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "flighttask_recovery"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: flighttask_recovery

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "need_reply": 1,
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "flighttask_recovery"
}
```

航线暂停

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: flighttask_pause

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {},
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "flighttask_pause"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: flighttask_pause

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "need_reply": 1,
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "flighttask_pause"
}
```

取消任务

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: flighttask_undo

Column	Name	Type	constraint	Description
flight_ids	计划 ID	array	{"size": -, "item_type": text}	

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "flight_ids": [
      "aaaaaaa",
      "bbbbbbb"
    ]
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1234567890123,
  "method": "flighttask_undo"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: flighttask_undo

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1234567890123,
  "method": "flighttask_undo"
}
```

下发任务

若用户的云服务无法访问外网，需实现 [配置更新](#) 功能，下发可被云服务访问的 NTP 服务的 URL，以实现时钟同步，否则航线任务将无法正常运行。

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: flighttask_prepare

Data:

Column	Name	Type	constraint
flight_id	计划ID	text	
execute_time	开始执行时间	int	{"length":13}
task_type	任务类型	enum_int	{"0":"立即任务","1":"定时任务","件任务"}
file	航线文件对象	struct	
»url	文件URL	text	
»fingerprint	文件签名	text	

ready_conditions	任务就绪条件	struct	
»battery_capacity	电池容量	int	
»begin_time	任务可执行时段 的开始时间	int	{"length":13}
»end_time	任务可执行时段 的结束时间	int	{"length":13}
executable_conditions	任务执行条件	struct	
»storage_capacity	存储容量	int	
break_point	航线断点信息	struct	

»index	断点序号	int	
»state	断点状态	enum_int	{"0":"在航段上","1":"在航点上"}
»progress	当前航段进度	float	{"max":1.0,"min":0}
»wayline_id	航线ID	int	
rth_altitude	返航高度	int	{"max":1500,"min":20,"unit_name":"/m"}
rth_mode	返航高度模式	enum_int	{"0":"智能高度","1":"设定高度"}
out_of_control_action	遥控器失控动作	enum_int	{"0":"返航","1":"悬停","2":"降落"}
exit_wayline_when_rc_lost	航线失控动作	enum_int	{"0":"继续执行航线任务","1":"退线任务，执行遥控器失控动作"}
wayline_precision_type	航线精度类型	enum_int	{"0":"GPS 任务","1":"高精度 RTK 任务"}

simulate_mission	是否在模拟器中执行任务	struct	
»is_enable	是否开启模拟器任务	enum_int	{"0":"不开启","1":"开启"}
»latitude	纬度	double	{"max":"90.0","min":"-90.0"}
»longitude	经度	double	{"max":"180.0","min":"-180.0"}
flight_safety_advance_check	飞行安全预检查	bool	{"0":"关闭","1":"开启"}

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "break_point": {
      "index": 1,
      "progress": 0.34,
      "state": 0,
      "wayline_id": 0
    },
    "executable_conditions": {
      "storage_capacity": 1000
    },
    "execute_time": 1234567890123,
    "exit_wayline_when_rc_lost": 0,

```

```

    },
    "flight_id": "xxxxxxx",
    "flight_safety_advance_check": 1,
    "out_of_control_action": 0,
    "ready_conditions": {
      "battery_capacity": 90,
      "begin_time": 1234567890123,
      "end_time": 1234567890123
    },
    "rth_altitude": 100,
    "simulate_mission": {
      "is_enable": 1,
      "latitude": 22.1223,
      "longitude": 113.2222
    },
    },
    "task_type": 2,
    "wayline_precision_type": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1234567890123,
  "method": "flighttask_prepare"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: flighttask_prepare

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1234567890123,
  "method": "flighttask_prepare"
}

```

创建航线任务（废弃）

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: flighttask_create

Data:

Column	Name	Type	constraint	Description
flight_id	计划 ID	text		
type	任务类型	text		
file	航线文件对象	struct		
»url	文件 URL	text		
»sign	MD5 签名	text		

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "file": {
      "sign": "xxxx",
      "url": "https://xxx.com/xxxx"
    },
    "flight_id": "xxxxxxxx",
    "type": "wayline"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "flighttask_create"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: flighttask_create

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "flighttask_create"
}

```

空中下发航线

飞行器处于空中飞行时，可以下发一些文件体积较小的航线

Method: `in_flight_wayline_deliver`

Topic: `thing/product/{gateway_sn}/services`

Direction: down

Data:

Column	Name	Type	Constraint	Descr
bid	请求标识	string		请求的唯一标识
tid	任务标识	string		任务的唯一标识
timestamp	时间戳	int64		请求的时间戳
method	方法名	string		固定为 <code>in_flight_w</code>
data	数据	object		下发航线的具体

data.file	文件信息	object		航线文件信息
data.file.fingerprint	文件指纹	string		用于校验文件完
data.file.url	文件下载地址	string		航线文件的下载
data.in_flight_wayline_id	航线任务ID	string		空中航线任务的
data.out_of_control_action	失控动作	int	{0: "返航", 1: "悬停", 2: "降落"}	遥控器失控时的
data.exit_wayline_when_rc_lost	失联动作	int	{0: "继续执行", 1: "执行飞控失控动作"}	航线失联时的动
data.rth_altitude	返航高度	int	{min: 20, max: 1500}	返航高度，单位
data.rth_mode	返航模式	int	{0: "智能高度", 1: "设定高度"}	返航高度模式
data.wayline_precision_type	航线精度类型	int	{0: "GPS任务", 1: "高精度RTK任务"}	航线精度类型

Example:

```

{
  "bid": "d26db6b4-3bb0-432c-bdb2-a8e53c827b72",
  "tid": "a5d95abe-e99c-4931-95cc-e2f3fdbbc488f",
  "timestamp": 1731135221528,
  "method": "in_flight_wayline_deliver",
  "data": {
    "file": {
      "fingerprint": "7a7cea5060c55920b7619a2a981f2223",
      "url": "XXXX"
    },
    "in_flight_wayline_id": "f2b31c13-3fce-4a7e-b188-87a78ff2f8a6",

```

```

        "rth_mode": 1,
        "wayline_precision_type": 1
    }
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Data:

Column	Name	Type	Constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
    "result": 0
}

```

暂停空中航线

Method: in_flight_wayline_stop

Topic: thing/product/{gateway_sn}/services

Direction: down

Data:

Column	Name	Type	Constraint	Description
bid	请求标识	string		请求的唯一标识符
tid	任务标识	string		任务的唯一标识符
timestamp	时间戳	int64		请求的时间戳

method	方法名	string		固定为 <code>in_flight_wayline</code>
data	数据	object		暂停航线的具体数据
data.in_flight_wayline_id	航线任务ID	string		空中航线任务的唯一标识

Example:

```

{
  "bid": "bf1d9985-5077-4ea5-9f2b-44edbcde9a11",
  "tid": "9da675b5-0a87-4b89-9813-4fb2e3aac4ec",
  "timestamp": 1731136137237,
  "method": "in_flight_wayline_stop",
  "data": {
    "in_flight_wayline_id": "36747381-4886-4ed0-9b2b-2919d6d4863c"
  }
}
```

Topic: `thing/product/{gateway_sn}/services_reply`

Direction: up

Data:

Column	Name	Type	Constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "result": 0
}
```

恢复空中航线

Method: `in_flight_wayline_recover`

Direction: down

Data:

Column	Name	Type	Constraint	Description
bid	请求标识	string		请求的唯一标识符
tid	任务标识	string		任务的唯一标识符
timestamp	时间戳	int64		请求的时间戳
method	方法名	string		固定为 <code>in_flight_wayline</code>
data	数据	object		恢复航线的具体数据
data.in_flight_wayline_id	航线任务ID	string		空中航线任务的唯一标识

Example:

```

{
  "bid": "8fd307b9-6a70-465c-aeb7-5edac860017b",
  "tid": "b3a76306-2576-4bc5-9a55-77e4f8f3b17c",
  "timestamp": 1731136244583,
  "method": "in_flight_wayline_recover",
  "data": {
    "in_flight_wayline_id": "36747381-4886-4ed0-9b2b-2919d6d4863c"
  }
}

```

Topic: `thing/product/{gateway_sn}/services_reply`

Direction: up

Data:

result	返回码	int	非 0 代表错误
--------	-----	-----	----------

Example:

```
{
  "result": 0
}
```

取消空中航线

Method: `in_flight_wayline_cancel`

Topic: `thing/product/{gateway_sn}/services`

Direction: down

Data:

Column	Name	Type	Constraint	Description
bid	请求标识	string		请求的唯一标识符
tid	任务标识	string		任务的唯一标识符
timestamp	时间戳	int64		请求的时间戳
method	方法名	string		固定为 <code>in_flight_wayline_cancel</code>
data	数据	object		取消航线的具体数据

Example:

```
{
  "bid": "8fd307b9-6a70-465c-aeb7-5edac860017b",
  "tid": "b3a76306-2576-4bc5-9a55-77e4f8f3b17c",
  "timestamp": 1731136244583,
  "method": "in_flight_wayline_cancel",
  "data": {}
}
```

Direction: up

Data:

Column	Name	Type	Constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{  
  "result": 0  
}
```

Requests

机场任务状态获取

蛙跳任务中，机场需要请求另一机场的任务执行状态

Topic: thing/product/{gateway_sn}/requests

Direction: up

Method: flighttask_progress_get

Data:

Column	Name	Type	constraint	Description
target_sn	目标设备sn	text		
flight_id	目标航线任务uuid	text		

Example:

```
{  
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "data": {  
    "flight_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
    "target_sn": "xxx"  
  },  
}
```

```
    "timestamp": 1234567890123
  }
```

Topic: thing/product/{gateway_sn}/requests_reply

Direction: down

Method: flighttask_progress_get

Data:

Column	Name	Type	constraint
result	返回码	int	
output	输出	struct	
»flight_id	设备最新当前任务id	text	
»progress	任务进度	struct	
»»current_step	执行步骤	int	
»»percent	执行进度	int	{"max":100,"min":0,"step":"","unit_name":null
»status	任务状态	text	

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",

```

```

        "progress": {
            "current_step": 27,
            "percent": 90
        },
        "status": "in_progress"
    },
    "result": 0
},
"gateway": "xxx",
"method": "flighttask_progress_get",
"tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
"timestamp": 0
}

```

任务资源获取

Topic: thing/product/{gateway_sn}/requests

Direction: up

Method: flighttask_resource_get

Data:

Column	Name	Type	constraint	Description
flight_id	计划 ID	text		

Example:

```

{
    "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "data": {
        "flight_id": "xxxxxxx"
    },
    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "timestamp": 1234567890123,
    "method": "flighttask_resource_get"
}

```

Topic: thing/product/{gateway_sn}/requests_reply

Direction: down

Method: flighttask_resource_get

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误
output	输出	struct		
»file	航线文件对象	struct		
»»url	文件 URL	text		
»»fingerprint	文件签名	text		文件 MD5 签名

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "output": {
      "file": {
        "fingerprint": "signxxxx",
        "url": "https://xx.oss-cn-hangzhou.aliyuncs.com/xx.kmz?Ex
      }
    },
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1234567890123,
  "method": "flighttask_resource_get"
}

```